SOFTWARE TESTING PRACTICAL REPORT

TUTORIAL – 12
AI-Powered Software Testing

Generative AI-Assisted Login Requirement Analysis and Selenium Automation

Student Name	Harshita Malviya
Student ID	2504030100426
Practical	Tutorial 12 — AI-Powered Software Testing
Application Under Test	SauceDemo — https://www.saucedemo.com/
Tools	Generative AI, Python, Selenium WebDriver, Google Chrome, PyCharm
Module	Login and Authentication

Prepared from the supplied Tutorial 12 lab manual, Python script, and execution screenshots.

1. Aim

To analyze AI-powered software testing and apply Generative AI techniques for requirement analysis, test-case generation, test-data creation, defect-prone area identification, Selenium script generation, code review, execution, and regression test planning.

2. Practical Overview

The practical focuses on a login requirement: a registered user should be able to log in with valid credentials, while invalid credentials should be rejected and an appropriate error message should be displayed. Generative AI is used as an assistant; its suggestions and generated code must be reviewed before execution.

3. Test Environment and Configuration

Parameter	Details
Project	AI-Powered Software Testing
Application	SauceDemo
Module	Login and Authentication
Programming Language	Python 3.x
IDE	PyCharm
Automation Tool	Selenium WebDriver
Browser	Google Chrome
Wait Strategy	WebDriverWait with 60-second timeout in the submitted script
Test Data	standard_user / secret_sauce

4. Requirement Analysis

Requirement	Testable Scenario
Valid user can log in	Positive login test
Invalid credentials are rejected	Negative login tests for wrong password and wrong username
An error message is shown	Error-message validation after invalid login
Username and password are required	Empty username and empty password tests
Existing login must keep working	Regression test for valid login

5. AI-Generated Test Cases

Test Case ID	Scenario	Test Data	Expected Result
TC-12-01	Valid Login	standard_user / secret_sauce	Products page opens
TC-12-02	Invalid Password	standard_user / wrong123	Error message displayed
TC-12-03	Invalid Username	wrong_user / secret_sauce	Error message displayed
TC-12-04	Empty Username	Blank / secret_sauce	Validation message displayed
TC-12-05	Empty Password	standard_user / Blank	Validation message displayed

6. Automation Script Under Test

The supplied script opens SauceDemo, waits for the username field, enters valid credentials, clicks Login, waits for the URL to contain “inventory”, checks the URL assertion, prints PASS messages, pauses for 15 seconds to allow screenshots, catches AssertionError, and closes the browser in a finally block.

Key validation points

• Explicit wait is used before entering the username.

• The URL assertion checks that the user reaches the inventory/products page.

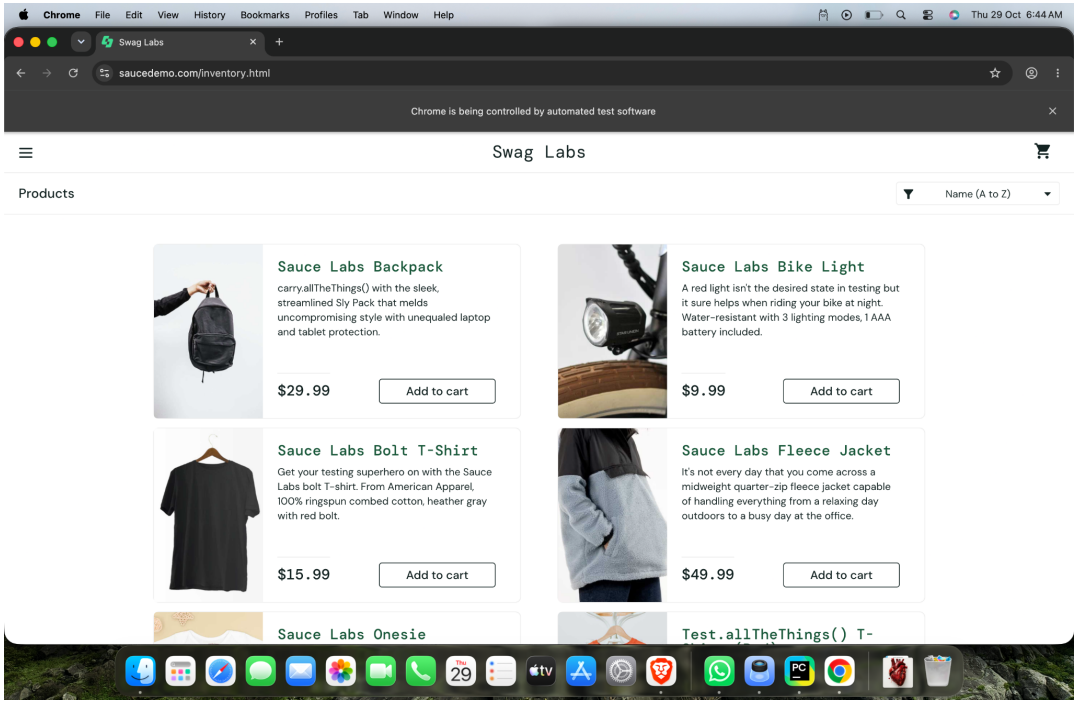
• The console prints “AI-Assisted Login Test: PASS” and “Requirement Validation: PASS” after the assertion succeeds.

• The finally block closes Chrome even if an assertion fails.

7. Execution Evidence

Screenshot A shows the SauceDemo inventory page at saucedemo.com/inventory.html, supporting that the login flow reached the products page. Screenshot B shows the PyCharm console output with “AI-Assisted Login Test: PASS”, “Requirement Validation: PASS”, and exit code 0.

Screenshot 1 — Browser after login

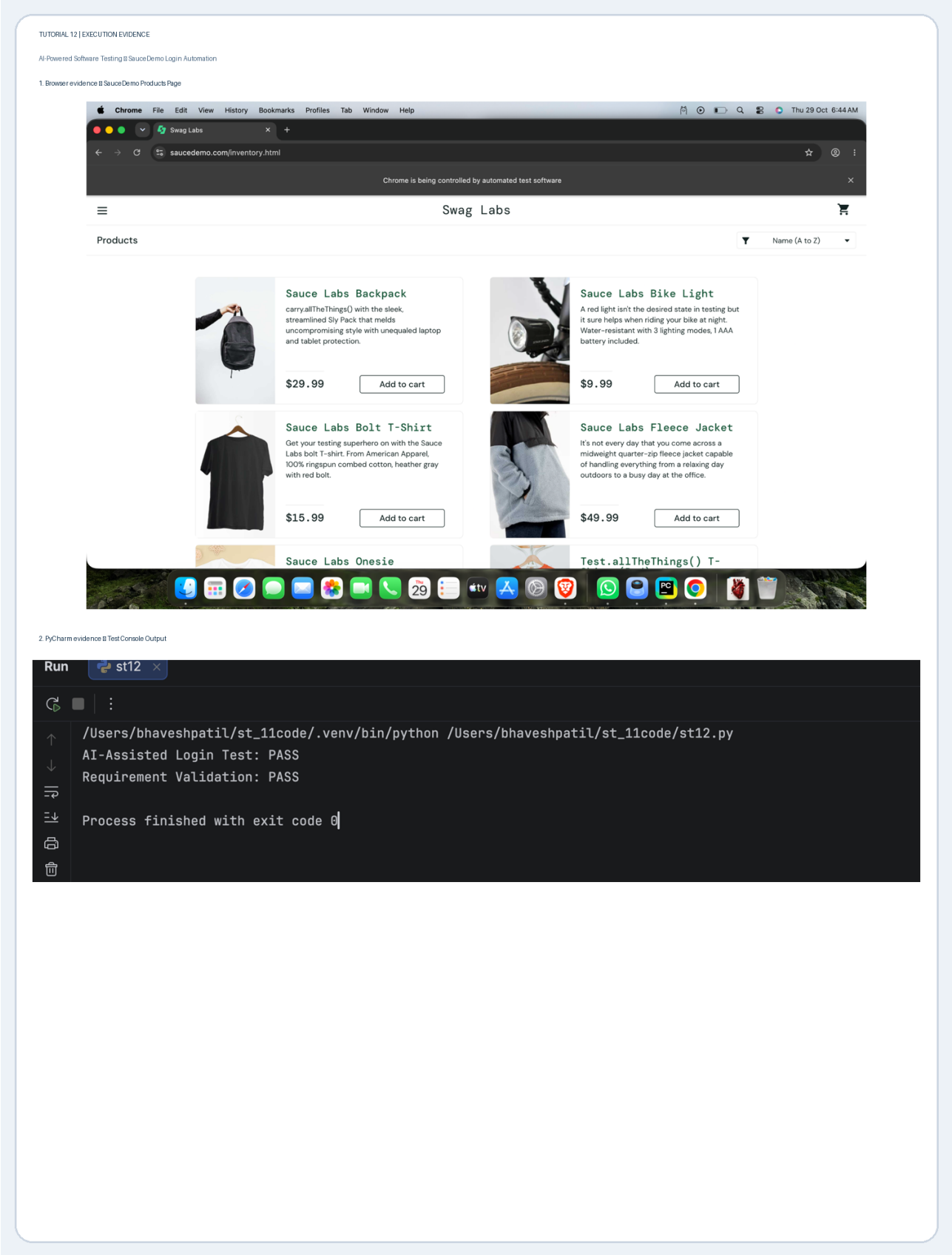


Screenshot 2 — PyCharm execution output



8. Screenshot Collage

The collage combines the supplied browser and console screenshots into a single evidence sheet.



9. Execution Summary

Metric	Observed Result
Automation script shown	1
Console login test result	PASS
Requirement validation	PASS
Python process exit code	0
Browser result	Products / inventory page visible

10. Defect-Prone Areas and Regression Recommendations

• Input validation: verify how the application handles blank username and password fields.

• Authentication: confirm invalid usernames and passwords are rejected.

• Error handling: confirm the correct error message is visible and understandable.

• Session/navigation: ensure a successful login reaches the inventory page and does not leave the user on the login page.

• Regression: repeat valid login after changes and re-run invalid-password, invalid-username, and empty-username tests.

11. Result and Conclusion

The supplied execution evidence indicates that the valid-login automation reached the SauceDemo inventory page and that the console validation passed with exit code 0. This practical demonstrates how Generative AI can assist with requirement analysis and test design, while Selenium WebDriver automates browser interactions. The supplied script tests the positive login path; the negative test cases listed in the manual are planned scenarios and are not shown as executed by this specific script.

12. Professional QA Notes

• The current script validates the positive login path only; add separate automated tests for invalid credentials and empty fields before reporting those cases as executed.

• For stronger verification, assert a visible Products heading or another inventory-page element in addition to checking the URL.

• The AssertionError handler prints a failure message but could also capture a screenshot and report the failed assertion details.

• AI-generated code and test cases should be reviewed by a human tester before use, as stated in the supplied lab manual.